Prompting Techniques

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1 Zero-Shot Prompting

Prompt:

Hi

Output:

Hello

Description:
Zero-shot prompting means giving the AI a task without providing any examples. The AI has to understand and perform the task based only on the instruction.

Main Difference: No examples are provided.

2 One-Shot Prompting

Prompt:

Example: Hello → Bonjour
Translate: Goodbye

Output:

Goodbye → Au revoir

Description:
One-shot prompting provides one example before asking the AI to perform a similar task. The example helps the AI understand the expected pattern.

Main Difference: Exactly one example is provided.

3 Few-Shot Prompting

Prompt:

Example 1: Hello → Bonjour
Example 2: Thanks → Merci
Translate: Goodbye

Output:

Goodbye → Au revoir

Description:
Few-shot prompting provides multiple examples to help the AI understand the pattern and produce the expected type of response.

Main Difference: Multiple examples are provided.

4 Role-Based Prompting

Prompt:

You are a senior doctor.
Explain fever in simple terms.

Output:

Fever means that your body temperature has increased, usually because your immune system is responding to an infection or another cause.

Description:
Role-based prompting assigns a specific role or persona to the AI. This influences the style, perspective, and level of explanation used in the response.

Main Difference: The AI is given a specific role or identity to follow.

5 Instruction-Based Prompting

Prompt:

Summarize the water cycle in 2 bullet points, formal tone.

Output:

- Water evaporates from Earth's surface, condenses into clouds, and forms precipitation.

- Precipitation returns water to rivers, lakes, oceans, and groundwater, completing the cycle.

Description:
Instruction-based prompting gives clear and specific instructions about what the AI should do. Instructions can specify the format, length, tone, or other requirements.

Main Difference: The focus is on clearly defining the task and requirements.

6 Chain-of-Thought (CoT) Prompting

Prompt:

> If a train travels 60 km in 1.5 hours, what is its speed?
> Let's think step by step.

Output:

> Speed = Distance ÷ Time
> Speed = 60 ÷ 1.5
> Speed = 40 km/h
>
> Therefore, the train's speed is 40 km/h.

Description:
Chain-of-thought prompting encourages the model to work through a problem in a sequence of reasoning steps before reaching the answer. It is commonly used for mathematical and logical problems.

Main Difference: The prompt encourages step-by-step reasoning rather than only asking for the final answer.

7 Structured Output / JSON Prompting

Prompt:

> Extract items and quantities in JSON:
> I bought 2 bananas and 3 apples.

Output:

```
{
  "items": [
    {
      "item": "banana",
      "quantity": 2
    },
    {
      "item": "apple",
      "quantity": 3
    }
  ]
}
```

Description:
Structured-output prompting tells the AI to return information in a specific format, such as JSON, XML, a table, or a predefined schema. This makes the output easier for programs to process.

Main Difference: The focus is on controlling the structure and format of the output.